

EXTENSION 1

Stage 2

Calculus (Ext1), C1 Rates of Change (Y11)

Rates of Change with respect to time (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 94.5 minutes (based on allocation of 1.5 minutes per mark)

### C1 Rates of Change



- Growth and Decay
- Related Rates of Change

**\*SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.**

### HISTORICAL CONTRIBUTION

- C1 Rates of Change* has contributed 4.3% per new syllabus Ext1 exam since it was introduced in 2020.
- This topic has been split into three sub-categories for analysis purposes which are: 1-*Growth and Decay* (2.6%), 2-*Rates of Change with respect to Time* (0%), and 3-*Related Rates of Change* (1.7%).
- This analysis looks at *Rates of Change with respect to Time*.

### HSC ANALYSIS - What to expect and common pitfalls

- Rates of Change with respect to Time* has not been examined in new syllabus Ext1 exams to date. It does however remain examinable content and should be revised.
- Rates of Change with respect to Time* is primarily reallocated content from the old *Mathematics* course. Past HSC questions on motion taken from this course dominate the question database and are identified by EXT1\* in the title.
- Identifying whether motion questions fit into the Adv or Ext1 course has been a challenge, although we note the use of  $\dot{x}$ ,  $\ddot{x}$  notation is specific to the Ext1 course. Our current view is that this area is examinable within both Advanced and Ext1, and the use of notation and the degree of difficulty will differentiate which course a question is most likely to be examined in.
- Flow questions are fair game within this topic area (see NESAs Calculus Topic Guide p.7) and we recommend a review of *Calculus, EXT1 C1 EQ-Bank 12* for exposure to this question type.

### Questions

#### 1. Calculus, EXT1\* C1 2019 HSC 10 MC

A particle is moving along a straight line with displacement  $x$  at time  $t$ .

The particle is stationary when  $t = 11$  and when  $t = 13$ .

Which of the following MUST be true in this case?

- A. The particle changes direction at some time between  $t = 11$  and  $t = 13$ .
- B.The displacement function of the particle has a stationary point at some time between  $t = 11$  and  $t = 13$ .
- C.The acceleration of the particle is 0 at some time between  $t = 11$  and  $t = 13$ .
- D.The acceleration function of the particle has a stationary point at some time between  $t = 11$  and  $t = 13$ .

#### 2. Calculus, EXT1\* C1 2013 HSC 10 MC

A particle is moving along the  $x$ -axis. The displacement of the particle at time  $t$  seconds is  $x$  metres.

At a certain time,  $\dot{x} = -3 \text{ ms}^{-1}$  and  $\ddot{x} = 2 \text{ ms}^{-2}$ .

Which statement describes the motion of the particle at that time?

- A. The particle is moving to the right with increasing speed.
- B. The particle is moving to the left with increasing speed.
- C. The particle is moving to the right with decreasing speed.
- D. The particle is moving to the left with decreasing speed.

#### 3. Calculus, EXT1 C1 EQ-Bank 12

A tank is initially full. It is drained so that at time  $t$  seconds the volume of water,  $V$ , in litres, is given by

$$V = 50\left(1 - \frac{t}{80}\right)^2 \text{ for } 0 \leq t \leq 100$$

- How much water was initially in the tank? **(1 mark)**
- After how many seconds was the tank one-quarter full? **(1 mark)**
- At what rate was the water draining out the tank when it was one-quarter full? **(2 marks)**

#### 4. Calculus, EXT1 C1 EQ-Bank 1

A particle is moving along the  $x$ -axis with a velocity,  $\dot{x}$ , in metres per second at  $t$  seconds, is given by the function

$$\dot{x} = \sqrt{5t + 4t^2 - t^3}$$

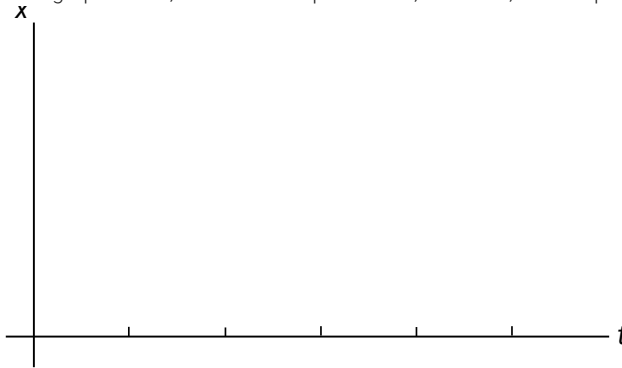
Find the acceleration of the particle when  $t = 3$ . **(2 marks)**

## 5. Calculus, 2ADV C1 EQ-Bank 8

The displacement  $x$  metres from the origin at time,  $t$  seconds, of a particle travelling in a straight line is given by

$$x = t^3 - 9t^2 + 9t, \quad t \geq 0$$

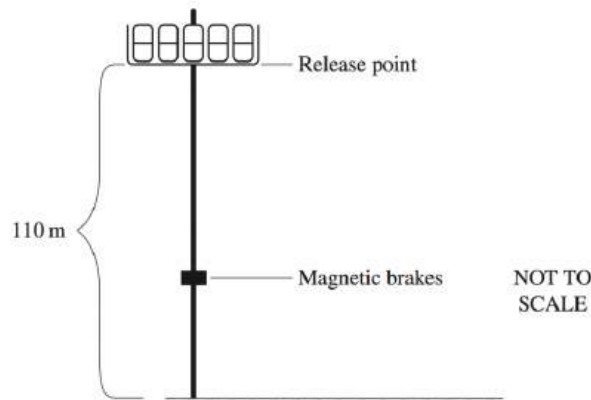
- Find the time(s) when the particle is at the origin. **(2 marks)**
- On the graph below, sketch the displacement,  $x$  metres, with respect to time  $t$ . **(2 marks)**



- Find the acceleration of the particle when  $t = 2$ . **(2 marks)**

## 6. Calculus, EXT1\* C1 2015 HSC 14a

In a theme park ride, a chair is released from a height of 110 metres and falls vertically. Magnetic brakes are applied when the velocity of the chair reaches  $-37$  metres per second.



The height of the chair at time  $t$  seconds is  $x$  metres. The acceleration of the chair is given by  $\ddot{x} = -10$ . At the release point,  $t = 0$ ,  $x = 110$  and  $\dot{x} = 0$ .

- Using calculus, show that  $x = -5t^2 + 110$ . **(2 marks)**
- How far has the chair fallen when the magnetic brakes are applied? **(2 marks)**

## 7. Calculus, EXT1 C1 EQ-Bank 2

A particle is moving along the  $x$ -axis where its displacement, in metres from the origin, after  $t$  seconds, is given by:

$$x = t^3 - 5t^2 + 8t - 6.$$

- Determine the times when the particle is at rest. **(2 marks)**
- Find and describe the velocity of the particle at the time where there is no accelerating force acting on it. **(2 marks)**

## 8. Calculus, EXT1\* C1 2013 HSC 14a

The velocity of a particle moving along the  $x$ -axis is given by  $\dot{x} = 10 - 2t$ , where  $x$  is the displacement from the origin in metres and  $t$  is the time in seconds. Initially the particle is 5 metres to the right of the origin.

- Show that the acceleration of the particle is constant. **(1 mark)**
- Find the time when the particle is at rest. **(1 mark)**
- Show that the position of the particle after 7 seconds is 26 metres to the right of the origin. **(2 marks)**
- Find the distance travelled by the particle during the first 7 seconds. **(2 marks)**

## 9. Calculus, EXT1\* C1 2010 HSC 7a

The acceleration of a particle is given by

$$\ddot{x} = 4\cos 2t,$$

where  $x$  is the displacement in metres and  $t$  is the time in seconds.

Initially the particle is at the origin with a velocity of  $1 \text{ ms}^{-1}$ .

- Show that the velocity of the particle is given by  $\dot{x} = 2\sin 2t + 1$ . **(2 marks)**
- Find the time when the particle first comes to rest. **(2 marks)**
- Find the displacement,  $x$ , of the particle in terms of  $t$ . **(2 marks)**

## 10. Calculus, EXT1\* C1 2005 HSC 9a

A particle is initially at rest at the origin. Its acceleration as a function of time,  $t$ , is given by

$$\ddot{x} = 4\sin 2t$$

- Show that the velocity of the particle is given by  $\dot{x} = 2 - 2\cos 2t$ . **(2 marks)**
- Sketch the graph of the velocity for  $0 \leq t \leq 2\pi$  AND determine the time at which the particle first comes to rest after  $t = 0$ . **(3 marks)**
- Find the distance travelled by the particle between  $t = 0$  and the time at which the particle first comes to rest after  $t = 0$ . **(2 marks)**

11. Calculus, EXT1\* C1 2007 HSC 5b

A particle is moving on the  $x$ -axis and is initially at the origin. Its velocity,  $v$  metres per second, at time  $t$  seconds is given by

$$v = \frac{2t}{16 + t^2}.$$

- i. What is the initial velocity of the particle? (1 mark)
- ii. Find an expression for the acceleration of the particle. (2 marks)
- iii. Find the time when the acceleration of the particle is zero. (1 mark)
- iv. Find the position of the particle when  $t = 4$ . (3 marks)

12. Calculus, EXT1\* C1 2017 HSC 15c

Two particles move along the  $x$ -axis.

When  $t = 0$ , particle  $P_1$  is at the origin and moving with velocity 3.

For  $t \geq 0$ , particle  $P_1$  has acceleration given by  $a_1 = 6t + e^{-t}$ .

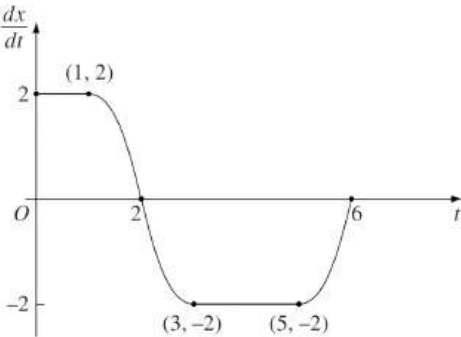
- i. Show that the velocity of particle  $P_1$  is given by  $v_1 = 3t^2 + 4 - e^{-t}$  (2 marks)

When  $t = 0$ , particle  $P_2$  is also at the origin.

For  $t \geq 0$ , particle  $P_2$  has velocity given by  $v_2 = 6t + 1 - e^{-t}$ .

- ii. When do the two particles have the same velocity? (2 marks)
- iii. Show that the two particles do not meet for  $t > 0$ . (3 marks)

13. Calculus, EXT1\* C1 2005 HSC 7b

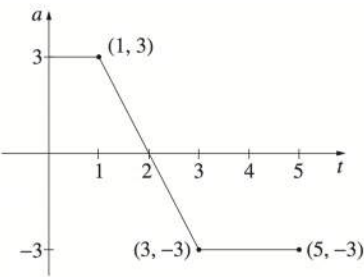


The graph shows the velocity,  $\frac{dx}{dt}$ , of a particle as a function of time. Initially the particle is at the origin.

- i. At what time is the displacement,  $x$ , from the origin a maximum? (1 mark)
- ii. At what time does the particle return to the origin? Justify your answer. (2 marks)
- iii. Draw a sketch of the acceleration,  $\frac{d^2x}{dt^2}$ , as a function of time for  $0 \leq t \leq 6$ . (2 marks)

14. Calculus, EXT1\* C1 2004 HSC 9b

A particle moves along the  $x$ -axis. Initially it is at rest at the origin. The graph shows the acceleration,  $a$ , of the particle as a function of time  $t$  for  $0 \leq t \leq 5$ .



- i. Write down the time at which the velocity of the particle is a maximum. (1 marks)
- ii. At what time during the interval  $0 \leq t \leq 5$  is the particle furthest from the origin? Give brief reasons for your answer. (2 marks)

## Worked Solutions

### 1. Calculus, EXT1\* C1 2019 HSC 10 MC

Given  $v = 0$  at  $t = 11$  and  $t = 13$ ,

Consider option  $C$ :

If  $v$  remains 0,  $a = 0$ .

If  $v$  increases ( $a > 0$ ) or decreases ( $a < 0$ ) after  $t = 11$ ,

it must then decrease ( $a < 0$ ) or increase ( $a > 0$ )

respectively so that  $v = 0$  when  $t = 13$ .

In each scenario,  $a = 0$  between  $t = 11$  and  $t = 13$ .

$\Rightarrow C$

◆ Mean mark 37%.

### 2. Calculus, EXT1\* C1 2013 HSC 10 MC

Since  $\dot{x} = -3\text{ms}^{-1}$

$\Rightarrow$  Particle is moving to the left

Since  $\ddot{x} = 2\text{ms}^{-2}$

$\Rightarrow$  Acceleration is to the right, against

the particle's motion

$\therefore$  Speed is decreasing

$\Rightarrow D$

◆◆ Mean mark 26%.

## Worked Solutions

### 3. Calculus, EXT1 C1 EQ-Bank 12

i. Initially,  $t = 0$ .

$$\begin{aligned} V &= 50(1-0)^2 \\ &= 50 \text{ L} \end{aligned}$$

ii. Find  $t$  when tank is  $\frac{1}{4}$  full:

$$\frac{50}{4} = 50 \left(1 - \frac{t}{80}\right)^2$$

$$\left(1 - \frac{t}{80}\right)^2 = \frac{1}{4}$$

$$1 - \frac{t}{80} = \frac{1}{2}$$

$$\frac{t}{80} = \frac{1}{2}$$

$$t = 40 \text{ seconds}$$

$$\begin{aligned} \text{iii. } \frac{dv}{dt} &= 2 \cdot 50 \left(1 - \frac{t}{80}\right) \left(-\frac{1}{80}\right) \\ &= -\frac{5}{4} \left(1 - \frac{t}{80}\right) \end{aligned}$$

When  $t = 40$ ,

$$\begin{aligned} \frac{dv}{dt} &= -\frac{5}{4} \left(1 - \frac{40}{80}\right) \\ &= -\frac{5}{8} \end{aligned}$$

$\therefore$  Water is draining out at  $\frac{5}{8}$  litres per second.

#### 4. Calculus, EXT1 C1 EQ-Bank 1

$$\dot{x} = (5t + 4t^2 - t^3)^{\frac{1}{2}}$$

$$\begin{aligned}\ddot{x} &= \frac{d}{dt}(\dot{x}) \\ &= \frac{1}{2}(5t + 4t^2 - t^3)^{-\frac{1}{2}} \times (5 + 8t - 3t^2)\end{aligned}$$

When  $t = 3$ ,

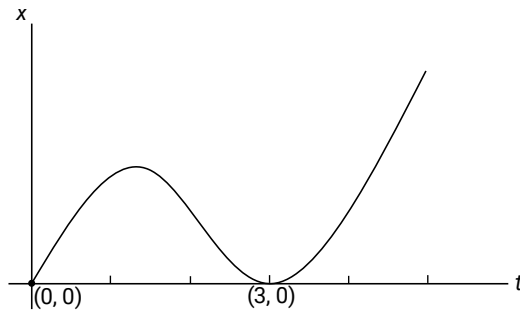
$$\begin{aligned}\ddot{x} &= \frac{1}{2}(15 + 36 - 27)^{-\frac{1}{2}} \times (5 + 24 - 27) \\ &= \frac{2}{2\sqrt{24}} \\ &= \frac{\sqrt{6}}{12} \text{ ms}^{-2}\end{aligned}$$

#### 5. Calculus, 2ADV C1 EQ-Bank 8

**a.** 
$$\begin{aligned}x &= t^3 - 9t^2 + 9t \\ &= t(t^2 - 9t + 9) \\ &= t(t - 3)^2\end{aligned}$$

Particle at origin when  $t = 0, t = 3$ .

**b.**



**c.** 
$$\begin{aligned}x &= t^3 - 9t^2 + 9t \\ v &= \frac{dx}{dt} = 3t^2 - 18t + 9 \\ a &= \frac{dv}{dt} = 6t - 18\end{aligned}$$

When  $t = 2$ :

$$a = 6 \times 2 - 18 = -6 \text{ m/s}^2$$

#### 6. Calculus, EXT1\* C1 2015 HSC 14a

**i.** Show  $x = -5t^2 + 110$

$$\ddot{x} = -10$$

$$\begin{aligned}\dot{x} &= \int \ddot{x} dt \\ &= \int -10 dt \\ &= -10t + c\end{aligned}$$

When  $t = 0$ ,  $\dot{x} = 0$

$$\begin{aligned}\therefore 0 &= -10(0) + c \\ c &= 0\end{aligned}$$

$$\dot{x} = -10t$$

$$\begin{aligned}x &= \int \dot{x} dt \\ &= \int -10t dt \\ &= -5t^2 + c\end{aligned}$$

When  $t = 0$ ,  $x = 110$

$$\begin{aligned}\therefore 110 &= -5(0^2) + c \\ c &= 110\end{aligned}$$

$$\therefore x = -5t^2 + 110 \text{ ... as required.}$$

**ii.** Find  $t$  when  $\dot{x} = -37$

$$\begin{aligned}-37 &= -10t \\ t &= 3.7 \text{ seconds}\end{aligned}$$

When  $t = 3.7$

$$\begin{aligned}x &= -5(3.7^2) + 110 \\ &= -68.45 + 110 \\ &= 41.55\end{aligned}$$

$$\begin{aligned}\therefore \text{Distance the chair has fallen} \\ &= 110 - 41.55 \\ &= 68.45 \text{ m}\end{aligned}$$

## 7. Calculus, EXT1 C1 EQ-Bank 2

i.  $x = t^3 - 5t^2 + 8t - 6$

$$\dot{x} = 3t^2 - 10t + 8$$

$$\ddot{x} = 6t - 10$$

Particle at rest  $\Rightarrow \dot{x} = 0$ :

$$3t^2 - 10t + 8 = 0$$

$$(3t - 4)(t - 2) = 0$$

$$t = \frac{4}{3} \text{ s or } t = 2 \text{ s}$$

ii. No acceleration  $\Rightarrow \ddot{x} = 0$

$$6t - 10 = 0 \Rightarrow t = \frac{5}{3} \text{ s}$$

Find  $\dot{x}$  when  $t = \frac{5}{3}$ :

$$\dot{x} = 3\left(\frac{5}{3}\right)^2 - 10\left(\frac{5}{3}\right) + 8$$

$$= \frac{25}{3} - \frac{50}{3} + \frac{24}{3}$$

$$= -\frac{1}{3} \text{ m/s}$$

$\therefore$  No accelerating force when  $\dot{x} = -\frac{1}{3} \text{ m/s}$  (towards the left)

## 8. Calculus, EXT1\* C1 2013 HSC 14a

i.  $\dot{x} = 10 - 2t$

$$\text{Acceleration} = \ddot{x} = \frac{d}{dt}\dot{x} = -2$$

$\therefore$  The acceleration is constant.

ii. Particle comes to rest when  $\dot{x} = 0$

$$10 - 2t = 0 \text{ when } t = 5$$

$\therefore$  Particle comes to rest after 5 seconds.

iii. Show  $x = 26$  when  $t = 7$

$$x = \int \dot{x} dt$$

$$= \int (10 - 2t) dt$$

$$= 10t - t^2 + c$$

Given  $t = 0$  when  $x = 5$

$$\Rightarrow 5 = 10(0) - 0^2 + c$$

$$c = 5$$

$$\therefore x = 10t - t^2 + 5$$

At  $t = 7$

$$x = 10(7) - 7^2 + 5$$

$$= 70 - 49 + 5$$

$$= 26$$

$\therefore$  The particle is 26 metres to the right of the origin after 7 seconds ... as required

iv. Find the distance travelled in the first 7 seconds:

At  $t = 5$ ,

$$x = 10(5) - 5^2 + 5$$

$$= 50 - 25 + 5$$

$$= 30 \text{ m}$$

$\Rightarrow$  The particle travels 25 m to the right ( $x = 5$  to 30)  
then 4 m to the left ( $x = 30$  to 26)

♦ Mean mark 37%.

**IMPORTANT:** Students can also draw a number line that shows where a particle starts from, changes direction and finishes in order to reduce errors.

$$\begin{aligned}
 \therefore \text{ The total distance travelled in 7 seconds} \\
 &= 25 + 4 \\
 &= 29 \text{ m}
 \end{aligned}$$


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## 9. Calculus, EXT1\* C1 2010 HSC 7a

i. Show  $\dot{x} = 2\sin 2t + 1$

$$\begin{aligned}
 \dot{x} &= \int \ddot{x} \, dt \\
 &= \int 4\cos 2t \, dt \\
 &= 2\sin 2t + c
 \end{aligned}$$

When  $t = 0$ ,  $\dot{x} = 1$  (given)

$$1 = 2\sin 0 + c$$

$$c = 1$$

$\therefore \dot{x} = 2\sin 2t + 1$  ... as required

ii. Find  $t$  when  $\dot{x} = 0$ :

$$2\sin 2t + 1 = 0$$

$$\sin 2t = -\frac{1}{2}$$

$$\Rightarrow \sin \theta = \frac{1}{2} \text{ when } \theta = \frac{\pi}{6}$$

Since  $\sin \theta$  is negative in 3rd and 4th quadrants

$$2t = \pi + \frac{\pi}{6}$$

$$2t = \frac{7\pi}{6}$$

$$t = \frac{7\pi}{12}$$

$\therefore$  Particle first comes to rest at  $t = \frac{7\pi}{12}$  seconds

iii.  $x = \int \dot{x} \, dt$

$$= \int (2\sin 2t + 1) \, dt$$

$$= t - \cos 2t + c$$

When  $t = 0$ ,  $x = 0$  (given)

$$0 = 0 - \cos 0 + c$$

$$c = 1$$

$\therefore x = t - \cos 2t + 1$

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# 10. Calculus, EXT1\* C1 2005 HSC 9a

i. Show  $\dot{x} = 2 - 2 \cos 2t$

$$\ddot{x} = 4 \sin 2t$$

$$\dot{x} = \int \ddot{x} dt$$

$$= \int 4 \sin 2t dt$$

$$= -2 \cos 2t + c$$

When  $t = 0$ ,  $x = 0$

$$0 = -2 \cos 0 + c$$

$$c = 2$$

$\therefore \dot{x} = 2 - 2 \cos 2t$  ... as required

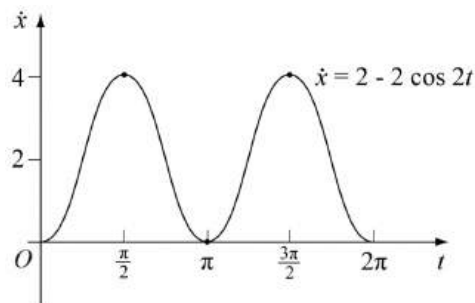
ii. Considering the range

$$-1 \leq \cos 2t \leq 1$$

$$-2 \leq 2 \cos 2t \leq 2$$

$$0 \leq 2 - 2 \cos 2t \leq 4$$

$$\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi$$



After  $t = 0$ , the particle next comes to rest at  $t = \pi$ .

iii. Distance travelled

$$= \int_0^\pi \dot{x} dt$$

$$= \int_0^\pi 2 - 2 \cos 2t dt$$

$$= [2t - \sin 2t]_0^\pi$$

$$= [(2\pi - \sin 2\pi) - (0 - \sin 0)]$$

$$= 2\pi \text{ units}$$

# 11. Calculus, EXT1\* C1 2007 HSC 5b

i.  $v = \frac{2t}{16 + t^2}$

When  $t = 0$

$$v = 0$$

$\therefore$  Initial velocity is 0.

ii.  $a = \frac{d}{dt} \left( \frac{2t}{16 + t^2} \right)$

Using quotient rule

$$u = 2t \quad v = 16 + t^2$$

$$u' = 2 \quad v' = 2t$$

$$\frac{dv}{dt} = \frac{u'v - uv'}{v^2}$$

$$= \frac{2(16 + t^2) - 2t \cdot 2t}{(16 + t^2)^2}$$

$$= \frac{32 + 2t^2 - 4t^2}{(16 + t^2)^2}$$

$$= \frac{2(16 - t^2)}{(16 + t^2)^2}$$

iii. Find  $t$  when  $\frac{dv}{dt} = 0$

$$\frac{2(16 - t^2)}{(16 + t^2)^2} = 0$$

$$2(16 - t^2) = 0$$

$$t^2 = 16$$

$$t = 4, t \geq 0$$

$\therefore$  The acceleration is zero when

$$t = 4 \text{ seconds.}$$

iv.  $v = \frac{2t}{16 + t^2}$

$$x = \int v dt$$

$$= \int \frac{2t}{16 + t^2}$$

$$= \log_e(16 + t^2) + c$$

When  $t = 0$  ,  $x = 0$

$$0 = \log_e(16 + 0) + c$$

$$c = -\log_e 16$$

$$\therefore x = \log_e(16 + t^2) - \log_e 16$$

When  $t = 4$ ,

$$x = \log_e(16 + 4^2) - \log_e 16$$

$$= \log_e 32 - \log_e 16$$

$$= \log_e \left( \frac{32}{16} \right)$$

$$= \log_e 2 \text{ metres}$$

$\therefore$  When  $t = 4$  , the position of the particle is  $\log_e 2$  metres.

## 12. Calculus, EXT1\* C1 2017 HSC 15c

i.  $a_1 = 6t + e^{-t}$

$$v_1 = \int a_1 \, dt$$

$$= \int 6t + e^{-t} \, dt$$

$$= 3t^2 - e^{-t} + c$$

When  $t = 0$ ,  $v_1 = 3$

$$3 = 0 - 1 + c$$

$$c = 4$$

$$\therefore v_1 = 3t^2 + 4 - e^{-t} \dots \text{as required}$$

ii.  $v_2 = 6t + 1 - e^{-t}$

Find  $t$  when  $v_1 = v_2$

$$3t^2 + 4 - e^{-t} = 6t + 1 - e^{-t}$$

$$3t^2 - 6t + 3 = 0$$

$$t^2 - 2t + 1 = 0$$

$$(t - 1)^2 = 0$$

$$\therefore t = 1$$

iii.  $x_1 = \int v_1 \, dt$

$$= \int 3t^2 + 4 - e^{-t} \, dt$$

$$= t^3 + 4t + e^{-t} + c$$

When  $t = 0$ ,  $x_1 = 0$

$$0 = 0 + 0 + 1 + c$$

$$c = -1$$

$$\therefore x_1 = t^3 + 4t + e^{-t} - 1$$

$$x_2 = \int 6t + 1 - e^{-t} \, dt$$

$$= 3t^2 + t + e^{-t} + c$$

When  $t = 0$ ,  $x_2 = 0$

$$0 = 0 + 0 + 1 + c$$

$$c = -1$$

$$\therefore x_2 = 3t^2 + t + e^{-t} - 1$$

◆ Mean mark (iii) 39%.

Find  $t$  when  $x_1 = x_2$

$$t^3 + 4t + e^{-t} - 1 = 3t^2 + t + e^{-t} - 1$$

$$t^3 - 3t^2 + 3t = 0$$

$$t(t^2 - 3t + 3) = 0$$

Since  $\Delta < 0$  for  $t^2 - 3t + 3$

$\Rightarrow$  No real solution

$\therefore$  The particles do not meet

$(x_1 \neq x_2)$  for  $t > 0$ .

### 13. Calculus, EXT1\* C1 2005 HSC 7b

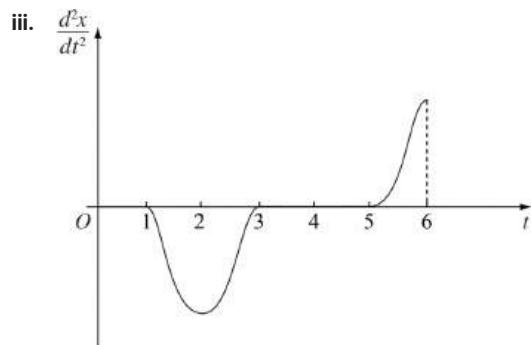
i. Maximum displacement in graph when

$$\frac{dx}{dt} = 0$$

$$\therefore t = 2$$

ii. Particle returns to the origin when  $t = 4$ .

The displacement can be calculated by the net area below the curve and since the area above the curve between  $t = 0$  and  $t = 2$  is equal to the area below the curve between  $t = 2$  and  $t = 4$ , the displacement returns to the initial displacement (i.e. the origin).



### 14. Calculus, EXT1\* C1 2004 HSC 9b

i. The velocity of the particle increases until

$$t = 2 \text{ (when } a = 0).$$

$\therefore$  Maximum velocity occurs at  $t = 2$ .

ii. It is furthest from the origin when its positive velocity

slows and becomes zero. This occurs when the “net” area under the curve is zero.

The area under the curve and above the  $x$ -axis between  $t = 0$  and 2, is equal to the area above

the curve and below the  $x$ -axis between  $t = 2$  and 4.

$\therefore$  Furthest from origin when  $t = 4$ .